

THE WATERWHEEL

A P R E P R I N T

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run ledger #20, #21, #24

3 seeds × 6 comparisons, zero reversals

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ABSTRACT

The brief imagined a reservoir transformed by, or living inside, the troughs of a water wheel. We built the wheel literally: the house 1024-unit reservoir's state is cut into n_{troughs} equal blocks, and every timestep exactly one trough is forcibly zeroed — under a **periodic** schedule (trough $t \bmod n$, a literal rotating wheel) or a **random** one (a uniformly random trough each step), the two dumping the identical total volume. Finding, replicated on three independently-drawn ponds with zero reversals in 18 registered comparisons: **the rotating schedule never costs more, and at heavier damage costs measurably less**. At coarse damage (1/8 of the state dumped per step) periodic beats random on ridge bpc, logistic bpc, *and* decode depth in every seed (logistic gap 0.0644–0.0661 bits); at fine damage (1/64) the schedules nearly tie (0.0053–0.0056) — a $\sim 12\times$ larger rhythm advantage under heavy damage, stable across all three seeds. Both schedules cost something against the undamaged pond: forced forgetting is never free. But a bounded worst-case staleness — no trough ever waits more than n steps, none is ever dumped twice in a row — is worth something that a text stream, which has no true period at 8 or 64 characters, "shouldn't" reward by a naive entropy argument.

Keywords reservoir computing · state resets · forgetting schedules · structured vs random perturbation · memory · pre-registration

§1 The question

Forced forgetting is usually treated as a volume question: how much state can you afford to lose per step? The wheel image sharpens it into a *schedule* question: holding the volume of forgetting exactly fixed, does the *rhythm* of it matter? A rotating wheel and a random dump destroy the same number of units per step; they differ only in structure — the wheel guarantees every trough a full circuit of accumulation time, while randomness occasionally leaves a trough untouched for a long drought or hits the same trough twice in a row. Random per-step zeroing of units is a familiar object in the training literature [2]; the wheel asks whether imposing deterministic structure on that zeroing, at matched volume, is worth anything at inference-time dynamics.

§2 The machine

The house champion reservoir [1] ($N=1024$, sparse tanh, $\rho=0.95$, fanin 10, house input map, leak 1.0 — wiring completely unchanged from the archived champion of the addressed-pond study) with one intervention: its units are cut into $n_troughs$ equal contiguous blocks, and immediately after each normal update, exactly one block is set to zero:

```
# periodic: trough (t mod n_troughs) dumps at step t
#   - bounded staleness: no trough waits > n_troughs steps,
#     none is ever double-dumped
# random:   a uniformly random trough dumps at step t
#   - identical total dump volume at matched n_troughs
x[trough] ← 0   # after the leaky-tanh update, every step
```

Two schedules $\times$ two trough counts ($n_troughs \in \{8, 64\}$): coarse damage, $1/8$ of the state per step; fine, $1/64 = 4$ cells per seed. The undamaged champion (ridge 3.1472, logistic 2.6727, depth 10.11) is cited from the archive, not rerun. Standard text8 splits [3], 2M train characters; instruments: ridge and logistic validation bpc, U3 decode depth.

§3 What we registered in advance

- **W1** (volume costs monotonically): coarse damage is worse than fine on all three instruments, at both schedules — 6/6.
- **W2** (manipulation check): all four wheel cells are worse than the undamaged pond on all three instruments — 12/12. Any forced dumping costs something.
- **W3** (the headline): periodic beats or ties random at both trough counts on all three instruments (tie bands ridge/logistic ± 0.005 , depth ± 0.2) — 6/6. Named alternative: any reversal means the wheel's regularity is actively harmful, e.g. a synchronized dump pattern colliding badly with the data's own structure.
- **W4** (the gap scales with damage): the periodic-vs-random logistic gap is larger at $1/8$ than at $1/64$.

Bands Q1–Q3 put numbers on the levels and the gaps. The registered follow-up policy promoted only W3, and only after replication; W1, W2, and W4 are supporting context even where they hit clean. The seed-1 rerun re-registered all four claims in the same wording; a third seed was then requested through the lab dashboard, re-registering the headline and gap claims (W3/W4) in identical wording — W1/W2 were supporting context, not re-registered a third time — with the pre-stated rule that a clean third hit appends a replication count and a reversal would be flagged rather than silently absorbed.

§4 What happened

Table 1. All three seeds (run ledger #20 / #21 / #24; values seed 0 / 1 / 2). Undamaged pond, cited: ridge 3.1472, logistic 2.6727, depth 10.11. Zero gate failures, zero AUTO-VOIDs.

cell	ridge val bpc	logistic val bpc	U3 depth
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periodic, 1/8	3.3712 / 3.3706 / 3.3740	2.8905 / 2.8937 / 2.8980	7.42 / 7.39 / 7.43
random, 1/8	3.4257 / 3.4249 / 3.4300	2.9564 / 2.9581 / 2.9641	7.06 / 7.02 / 7.06
periodic, 1/64	3.2718 / 3.2684 / 3.2756	2.7975 / 2.7950 / 2.8026	9.16 / 9.17 / 9.26
random, 1/64	3.2760 / 3.2723 / 3.2801	2.8031 / 2.8003 / 2.8080	9.14 / 9.13 / 9.21

W3, the headline: 6/6 in every seed, zero reversals ever. At coarse damage the wheel wins outright on all three instruments, every seed, well outside the tie bands. At fine damage the fine structure replicated, not just the direction: a ridge tie (gaps 0.0042/0.0039/0.0045, inside ± 0.005), a real logistic win just past the band (0.0056/0.0053/0.0054), and a depth tie — the same tie-win-tie pattern on all three independently-drawn ponds. W1 hit 6/6 and W2 hit 12/12 as registered on seeds 0 and 1; seed 2's table shows the same pattern in every comparison, though only W3/W4 carried registered bars there. Forced forgetting always costs something against the undamaged pond.

W4: the rhythm advantage scales with damage, stably. Periodic-vs-random logistic gaps: 0.0659 / 0.0644 / 0.0661 bits at 1/8 damage against 0.0056 / 0.0053 / 0.0054 at 1/64 — ratios 11.8 $\times$, 12.2 $\times$, 12.2 $\times$. The advantage of the wheel is $\sim 12\times$ larger under heavy damage, to within a percentage point, on three ponds that share no wiring and (in the random arms) no dump sequence.

Registered bands. At seed 0: Q1 hit; Q2 missed — the coarse wheel cost less than the smoke-informed band predicted (2.8905 against a floor of 2.95); Q3 missed at 1/8 and hit at 1/64 — the coarse rhythm advantage (0.0659) came in at more than double the band's upper end (0.03). Both misses run in the beat-expectations direction. The replication bands — held against the earlier seeds' actuals rather than the original conservative guesses — then hit 3/3 on both later seeds, with every bpc cell within a hundredth of a bit of its earlier-seed twin.

§5 What it means

A guaranteed turn beats a lottery. The stream being modeled has no true period at 8 or 64 characters, so a content-blind, input-independent dump schedule has no obvious phase structure to exploit — the same total information is destroyed either way. The schedule matters anyway. The plausible mechanism, recorded with the finding: worst-case staleness and double jeopardy. Random dumping occasionally leaves a trough undumped for many steps (its content grows stale, then is destroyed all at once) or dumps the same trough twice in a row (a wasted destruction); the rotation structurally forbids both. That risk bites harder when the per-step surviving budget is smaller — consistent with the measured $\sim 12\times$ scaling.

Forgetting is never free, but its schedule is a real design axis. Both schedules cost real bits and real depth against the undamaged pond at every setting — the wheel does not turn damage into profit. What it shows is that at fixed damage volume, *structure* in the forgetting schedule is worth up to 0.066 bits — roughly the size of effects this program treats as headline findings elsewhere — and that the worth is predictable from the damage rate. Follow-up axes named in the journal but not run: finer trough ladders, and non-contiguous trough partitions.

§6 References

1. H. Jaeger, "The 'echo state' approach to analysing and training recurrent neural networks," GMD Report 148, German National Research Center for Information Technology, 2001.
2. N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, R. Salakhutdinov, "Dropout: a simple way to prevent neural networks from overfitting," *Journal of Machine Learning Research* 15(56), 2014.
3. M. Mahoney, "Large text compression benchmark" (text8), mattmahoney.net/dc/textdata.

§7 Provenance

- **eb8cfoo8** — 2026-08-11-caity-waterwheel-in-the-reservoir: the brief; 4 cells, seed 0 (run ledger #20). Registered before running, smoke disclosed. Headline hit 6/6 ⇒ provisional, replication auto-queued before any findings entry.
- **ab4f82a2** — 2026-08-11-waterwheel-reservoir-rerun1: seed 1, fresh pond wiring and fresh random-dump sequence (run ledger #21). 6/6 again, zero reversals; promoted per the pre-stated rule.
- **79ae8b93** — 2026-08-11-caity-waterwheel-reservoir-rerun: seed 2, requested through the lab dashboard as a direct third-seed confirmation; the headline and gap claims re-registered in identical wording (run ledger #24). 6/6 a third time; findings annotated 3/3 seeds.

All claims judged strictly against the registered wording, misses included above; 3 seeds; replicated before publication.

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